

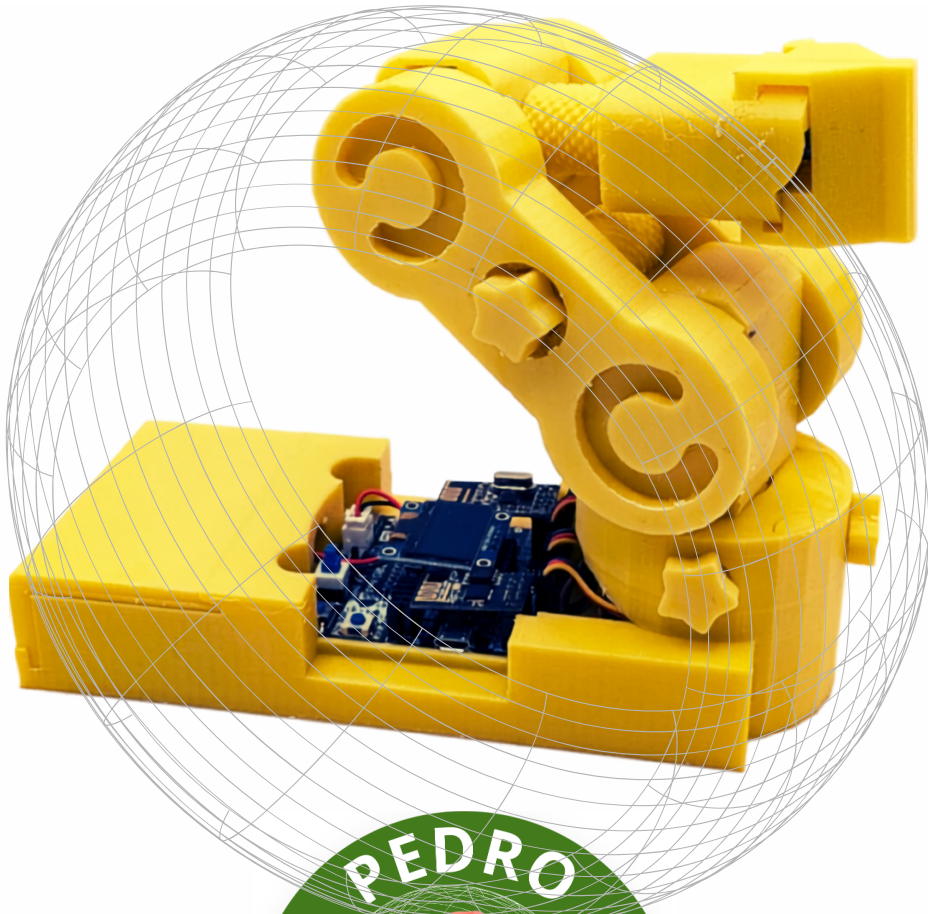
BEGINNER

pedro



STEM Program

Lesson #1: Pedro Discover



Complete this lesson
to earn your Pedro
Badge



Add it to your Pedro
Passport to track
your STEM journey

YOUR MISSION: Discover Pedro robot, the fully 3D Printing open source robot designed to make robotics learning fun and hands-on.

Welcome to the Pedro STEM Program

Throughout these **10 hands-on lessons**, you will discover the exciting world of **robotics, electronics, mechanics, and programming** by building your own Pedro robot, the fully 3D-printed open source robot designed to make robotics accessible for everyone.

During this learning journey, you will explore how robots:

-  **Think** through programming and decision-making.
-  **Move** using motors and mechanical systems.
-  **Communicate** using modern wired and wireless communication technologies.

As a complete robotics learning platform, Pedro supports multiple communication modes commonly used in robotics applications, including:

1. Manual Mode
2. Radio Mode
3. USB Serial Mode
4. Record & Replay Mode
5. UART Mode
6. USB Mode
7. Bluetooth Mode

Each lesson introduces a new concept through a progressive, hands-on approach. You will learn the underlying communication modes, configure the Pedro board, perform practical experiments, observe the robot's behavior, and complete STEM activities designed to reinforce your understanding.

At the end of every lesson, you will complete a **quiz** to test your knowledge and validate the skills you have learned.

- Lesson #1: Discover Pedro
- Lesson #2: The Brain of Pedro
- Lesson #3: Pedro Assembling
- Lesson #4: Manual Mode
- Lesson #5: Radio Mode
- Lesson #6: USB Serial Mode
- Lesson #7: Record & Replay Mode
- Lesson #8: UART Mode
- Lesson #9: Bluetooth Mode
- Lesson #10: Pedro Maker

Unlock Your Pedro Badges

At the end of each lesson, you will earn a **Pedro Badge** to recognize the new skills you have mastered.

After completing a lesson, **cut out (or receive) the corresponding Pedro Badge and stick it into your Pedro Passport**. This passport allows you to track your progress throughout the **Pedro STEM Journey** and celebrate each milestone you achieve. Each badge represents a new robotics skill, bringing you one step closer to becoming a **Pedro Robot Maker**.

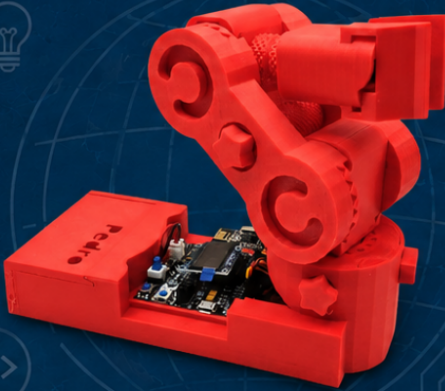


Collect all **10 Pedro Badges** to complete your Pedro Passport and successfully finish the Pedro STEM Program.

pedro

STEM PASSPORT

— Your Journey. Your Skills. Your Future. —



This Passport Belongs To:

★ Collect all 10 badges and become a Pedro Maker! ★

MY PEDRO BADGES

Complete each lesson to earn your badge!

1 PEDRO EXPLORER  Date: _____	2 PEDRO BOARD EXPERT  Date: _____	3 PEDRO BUILDER  Date: _____	4 PEDRO MOTION ENGINEER  Date: _____	5 PEDRO RADIO SPECIALIST  Date: _____
6 PEDRO AUTOMATION EXPERT  Date: _____	7 PEDRO BLUETOOTH ENGINEER  Date: _____	8 PEDRO SERIAL PROGRAMMER  Date: _____	9 PEDRO ROBOT DESIGNER  Date: _____	10 PEDRO MAKER  Date: _____



CONGRATULATIONS!

You completed all 10 lessons and unlocked all Pedro badges.

You are now an official

★ PEDRO MAKER! ★



Learning Goals

In this lesson, you will discover Pedro, the open-source educational robot designed to introduce robotics, electronics, programming, and engineering.

You will learn how Pedro is designed and how to identify its components.



🔑 At the end of this lesson, you will unlock your first Pedro Badge:
Pedro Discover

Teacher Notes

- No prior robotics experience required.
- Duration: 60–90 minutes
- Difficulty: ● Beginner
- Required Materials
 - 1 **fully Assembled** Pedro robot (*teacher demonstration models only*)
 - 1 **not Assembled** Pedro robot (*teacher demonstration models only*)
- **Important:** Do not distribute Pedro robots to students during this lesson. The robot is used exclusively by the teacher for demonstrations and classroom discussions.



Think: The **brain icon** encourages students to make careful observations, share their ideas, and engage in discussion by explaining and justifying their answers.



Learn: The **light bulb icon** indicates important concepts that students should read carefully. Students are encouraged to ask questions and discuss these topics about the robotics concepts presented in the lesson.



Make: The **pencil icon** indicates that students must complete the activity on the page. These activities are designed to reinforce the concepts learned through experimentation, and problem-solving.



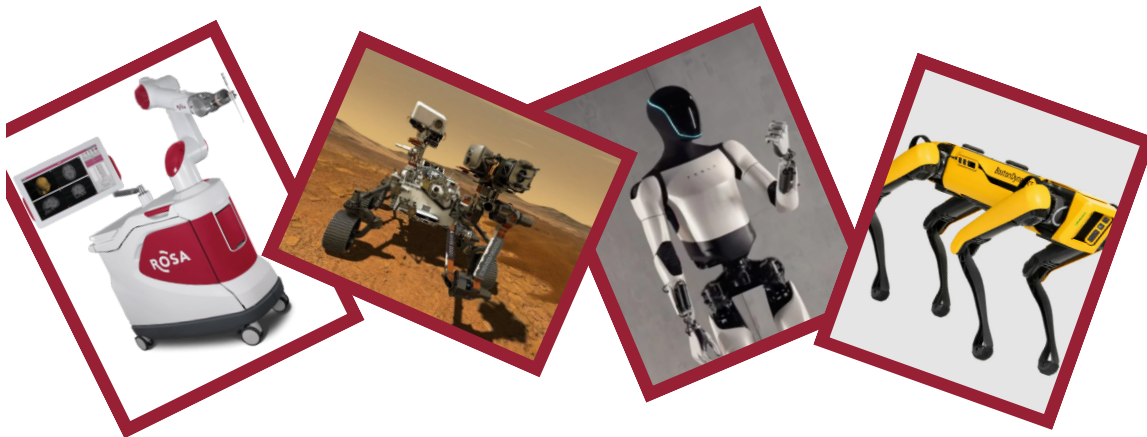
What is a Robot?

A robot is a machine that can perform tasks automatically or with human control. Robots help people by doing jobs that are dangerous, difficult, repetitive, or very precise. A robot usually has:

- 👁️ **Sensors** to sense the environment
- 🧠 **A controller** to process information
- ⚙️ **Motors and actuators** to move
- 🔋 **A power source** such as a battery or electricity

A Robot Can:

1. **Sense** the world using cameras, sensors, or microphones.
2. **Think** by processing information and making decisions.
3. **Act** by moving, picking up objects, speaking, or completing tasks.



Types of Robots:

- **Medical Robot:** Delivers medicine and supplies in hospitals.
- **Space Rover:** Explores Mars, takes photos, and collects rock samples.
- **Humanoid Robot:** A robot that looks like a person. It can walk, carry objects, and help people.
- **Quadruped Robot:** Walks on four legs and inspects dangerous places like factories and construction sites.



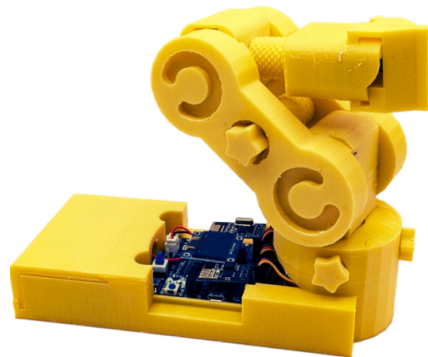
Did you ever see a robot in a real life? Which kind of robot was?
How could robots help people? Let students discuss in small groups.



Meet Pedro

P.E.D.R.O. (Programmable Educational Robot) is a fully 3D-printed, open-source robot designed to introduce students to the world of robotics, electronics, mechanics, and programming.

It is designed to teach the complete robotics design process, from mechanical assembly and electronic integration to programming and robot control. Thanks to its modular design, Pedro can be assembled **without any special tools**, making robotics more accessible for STEM education and hands-on learning activities.



**Tool-Free
Assembly**

As an open-source project, all the resources required to build, understand, and customize Pedro are freely available, including:

- 3D models and mechanical designs
- Electronic schematics
- Source code and software
- Technical documentation and tutorials

Pedro is officially certified by the **Open Source Hardware Association (OSHWA)** under **Certification ID: FR00027**, ensuring that its hardware design is openly shared and accessible to everyone.



You can find all Pedro resources here:

-  **Pedro Website:** <https://pedrobot.com>
-  **Pedro GitHub Repository:** <https://github.com/almtzr/Pedro>



Student Reflection

Observe and manipulate the Pedro robot with your classmates. Then discuss the following design concepts:

-  **Open Source**
-  **Fully 3D Printed**
-  **Tool-Free Assembly**

After observing the robot, answer the following questions:

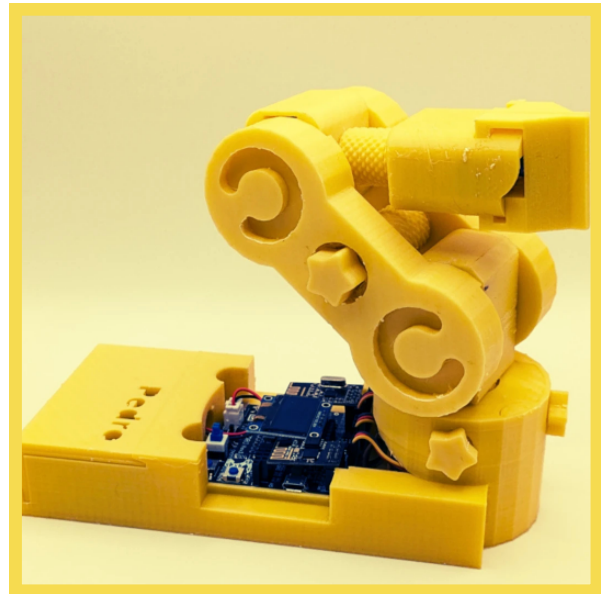
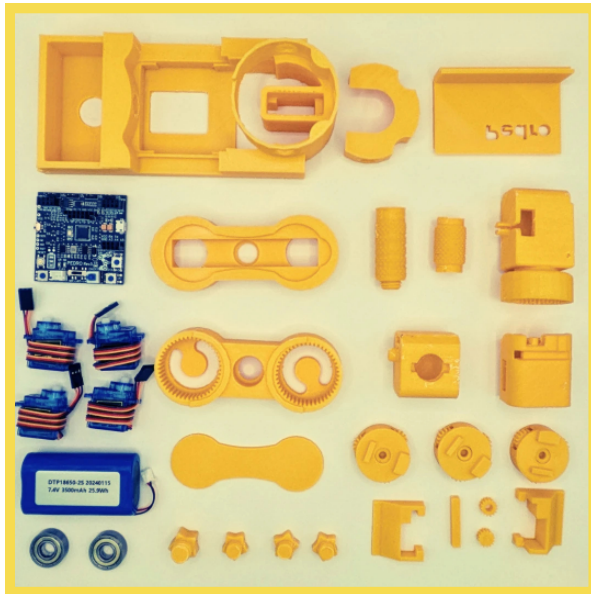
1. Why do you think an **open-source** robot is useful for education?

2. What are the advantages of a **fully 3D-printed** robot?

3. Why is **tool-free assembly** an advantage for a STEM activity?



Pedro Anatomy



3D-printed parts (The body)

Pedro is designed to be quick and easy assembly. Composed of **23 3D-printed parts** and **2 ball bearings** to help the servo motors rotate smoothly.

It can be assembled **without any tools**, making Pedro accessible to everyone. No prior experience or technical skills are required to assembling.



The actuators (The muscles)

Robots use **actuators** to produce movement. An actuator is a device that converts energy into mechanical motion.

In Pedro, the actuators are four **servo motors**, which provide the precise movements required to control the robot. Three servo motors control the movement axes, while the fourth servo motor is dedicated to the gripper.



The Control board (The brain)

Pedro is controlled by an **embedded electronic control board**, which acts as the **robot's brain** by processing information, making decisions, and controlling all of its movements and functions (more on this in the next lesson).

The control board is powered by a **7.4 V DC** rechargeable battery, allowing Pedro to operate anywhere without the constraints of power cables.



Student Reflection

1. Why is the electronic board often called the brain of Pedro?

2. What is the role of the ball bearings in Pedro?

3. Which component is responsible for creating Pedro's movements?

4. What features make Pedro easy to assemble?

5. Why are servo motors often called the muscles of a robot?

6. Why do you think Pedro is made from 3D-printed parts instead of metal parts?

7. Why does Pedro use a rechargeable battery instead of staying connected to a power cable?

8. What is the function of an actuator in a robot?



3D Printing (The Body)

Pedro is designed to be **fully manufacturable using a 3D printer**. Every structural part of the robot can be printed at home, in a makerspace, or at school.

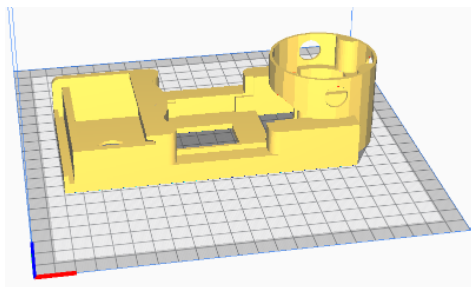
A 3D printer is a machine that creates real objects by printing one layer of plastic at a time. The two most common materials used for Pedro are:

- **PLA** – Easy to print and ideal for education.
- **ABS** – Stronger and more resistant.

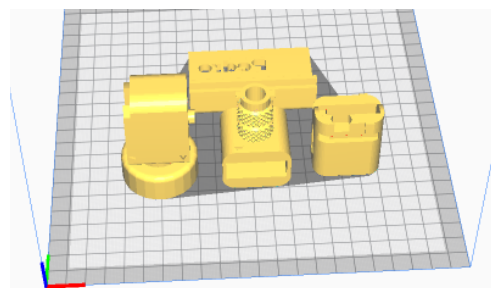
Before printing a part, several important steps are needed:

1. **Design** the part in CAD software.
2. **Export** the design as an STL file.
3. **Slice** the STL file to generate **G-code**.
4. **Print** the part using the G-code file in the 3D printer.

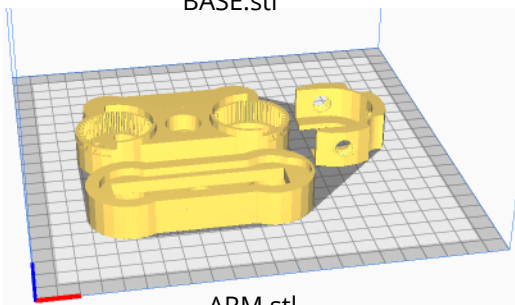
Pedro requires only four main printable models: **BASE.stl**, **SERVO.stl**, **ARM.stl**, and **GEAR.stl**. As an open-source project, all 3D printing files are available on the Pedro website: <https://pedrobot.com>.



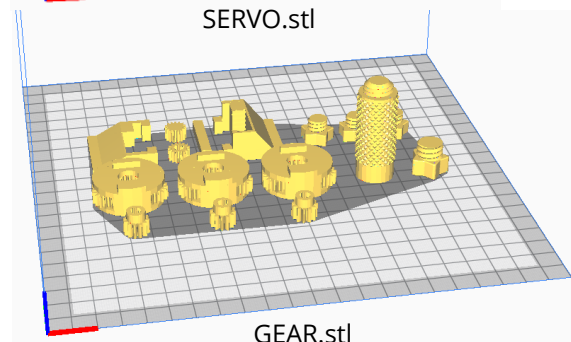
BASE.stl



SERVO.stl



ARM.stl



GEAR.stl



Student Reflection

1. What is a 3D printer? Have you ever seen or used one before?

2. Why is 3D printing an important technology for building Pedro?

3. Why must an STL file be converted into G-code before printing?

4. What is the difference between PLA and ABS?

5. Why do you think Pedro is made from only four main printable models?

6. What are the four main steps required to manufacture a Pedro part using a 3D printer?

7. Why is it useful that all Pedro 3D printing files are open source and freely available?

8. What is the role of CAD software before printing a part?



The Servo Motors (The Muscles)

Servo motors make Pedro move.

Servo motors are the **muscles** of the robot. They convert **electrical energy** from the battery into **mechanical movement**. Without servo motors, Pedro would never be able to move!



A servo motor is more than just a motor. Inside every servo motor are four main components:

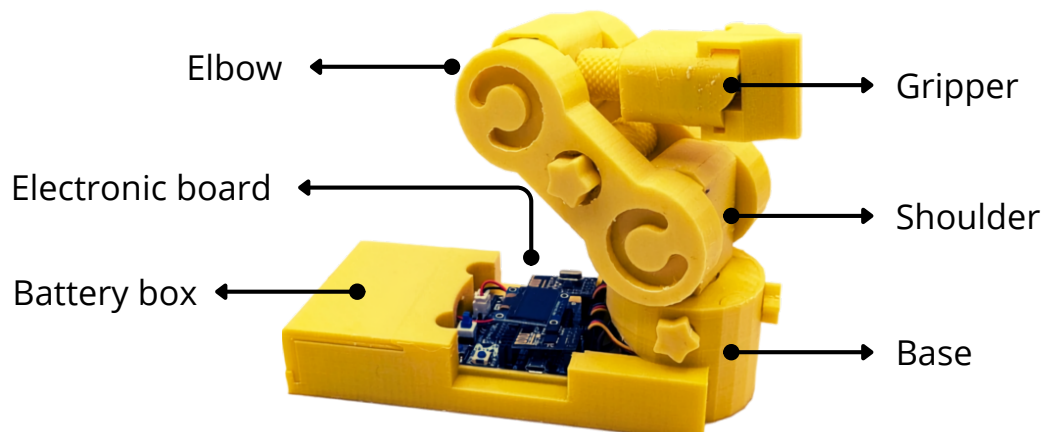
- ⚡ A **DC motor** that creates the movement.
- ⚙️ A **gearbox** that increases the motor's power (torque).
- 📡 A **sensor** that monitors the rotation.
- 🧠 An **electronic control circuit** that receives commands from Pedro.

Because of this built-in electronics, a servo motor can understand commands and move exactly as requested. **A servo motor is connected with three wires**

- **VCC (+5V):** Provides electrical power to the servo motor.
- **GND (Ground):** The electrical reference and completes the circuit.
- **SIG (Signal):** Receives signal to control the servo motor's movement.

Pedro is equipped with **four 360° continuous rotation servo motors**. Each servo controls one part of the robot:

- Servo 1 → **Base** for horizontal rotation.
- Servo 2 → **Shoulder** for enabling vertical rotation.
- Servo 3 → **Elbow** for facilitating pick-up movements.
- Servo 4 → **Gripper** for grabbing object.



Together, these four servo motors allow Pedro to perform all of its movements. Every movement follows the same sequence:

Battery → Electronic Board → Signal Wire → Servo Motor → Robot Movement



Students can manipulate a servo motor and look inside it through the transparent casing to identify the electronic control board and observe the different internal components.

This hands-on observation helps students understand how robotics combines mechanics, electronics, and computer control into a single system.

There are two main types of servo motors commonly used in educational robotics:

- Standard servo motors

A standard servo motor controls a specific angular position. For example 0° , 90° or 180° . The controller sends a command, and the servo moves to the requested position.

- 360° continuous rotation servo motors

Pedro uses 360° continuous rotation servo motors. Unlike standard servos, they are not controlled by position. They behave more like controlled DC motors. The command controls rotation direction and rotation speed.

Using 360° servo motors makes Pedro simpler to control because each motor already contains its own electronics. The main control board only needs to send commands instead of directly controlling the motor electronics.

A servo motor is a small example of an embedded system. It contains:

- electronic components,
- sensors,
- a control circuit,
- a motor.

The servo receives information from Pedro's main control board and executes the requested movement.

The main board decides what Pedro should do, and each servo motor manages its own rotation.



Student Reflection

Understanding the servo motor is essential because it shows how modern robots combine mechanical parts, electronics, and software into one intelligent system.

1. What is an actuator?

2. How many wires connect the servo motor? and specify each pin

3. What is the difference between a motor and a servo motor?

4. Why does the servo need gears?

5. Why is having electronics inside the servo useful?

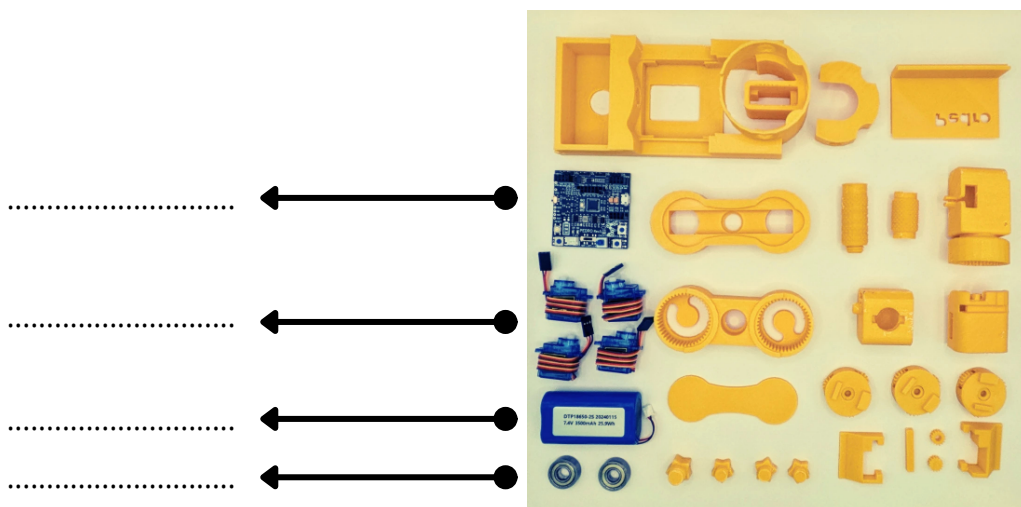
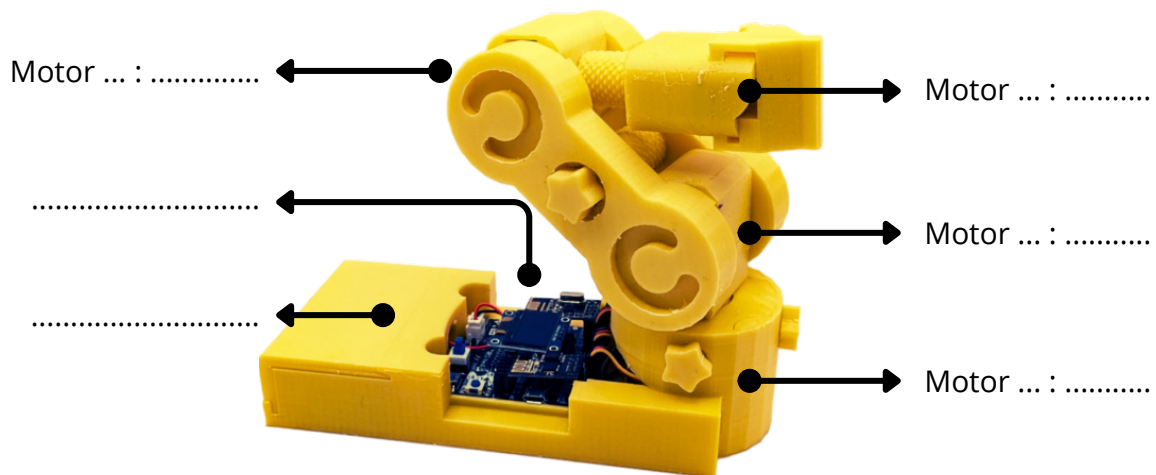


Components Identification

Look carefully at the pictures below and identify the different parts of Pedro.

- Match each servo motor with its position on the robot:
 - Base
 - Shoulder
 - Elbow
 - Gripper
- Identify the main components:
 - Electronic control board
 - Servo motors
 - Battery
 - Ball bearings

Explain the role of each component in Pedro's design.



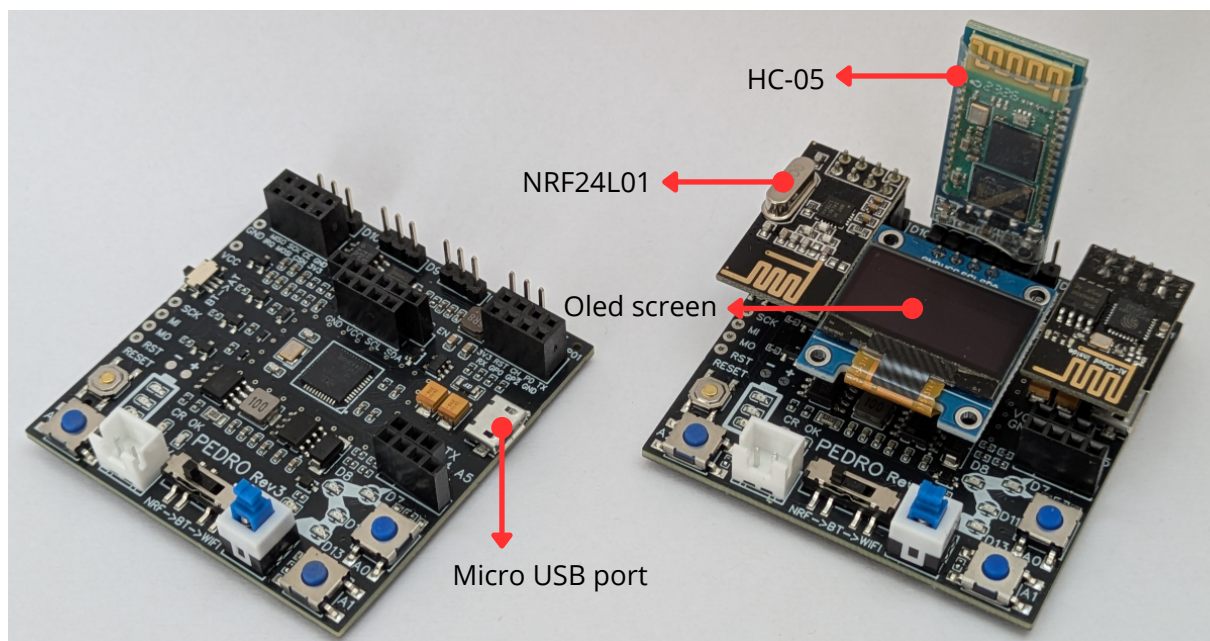


Multiple Control Modes

More than just a robot. Pedro is a complete educational robotics platform designed to discover how modern communication systems work.

It also provides six operating modes that allow students to understand how robots can be controlled in different ways:

1. 📶 **Radio Mode:** remote control using a radio module **NRF24L01**.
2. 📶 **Bluetooth Mode:** remote control via smartphone using module **HC-05**.
3. 🖱️ **USB Serial Mode:** control from a computer using a micro **USB cable**.
4. 🎮 **Normal Mode:** direct control using onboard buttons.
5. 🟡 **Record Mode:** record movements.
6. ▶️ **Replay Mode:** replay movements.
7. 🖱️ **UART Mode:** control by sending data through the **RX and TX pins**



Using an **128x64 OLED screen**, users can easily navigate through the menu to configure Pedro.

By using the same robot with different communication methods, students can clearly observe how information is transmitted, received, and processed.



Ask students to carefully observe Pedro before giving any explanations.

How do you think Pedro can be controlled?

How do you think a robot receives instructions?

Introduce the notion of remote communication



Quiz - Check Your Understanding

1. What does the acronym P.E.D.R.O. stand for?

- ☐ Programmable Electric Device for Robotic Operation
- ☐ Programmable Educational Robot
- ☐ Printed Electronic Design Robotic Object
- ☐ Personal Educational Digital Robot

2. What is a robot?

- ☐ A machine that can perform tasks automatically
- ☐ A type of computer game
- ☐ A human assistant
- ☐ A mobile phone

3. What component is the brain of a robot?

- ☐ Battery
- ☐ Motor
- ☐ Electronic board
- ☐ Gripper

4. What component allows the robot to move?

- ☐ Sensors
- ☐ Actuators / Motors
- ☐ Battery
- ☐ Frame

5. What kind of robot Pedro is?

- ☐ An humanoid robot
- ☐ A robotic arm
- ☐ A medical robot
- ☐ A space rover

6. Which statement is TRUE about Pedro?

- ☐ Pedro can only be controlled with one communication method
- ☐ Pedro supports Radio, Bluetooth, and USB Serial communication
- ☐ Pedro cannot record movements
- ☐ Pedro only works when connected to a computer

7. What material is used to print Pedro's parts?

- ☐ Aluminum
- ☐ PLA or ABS plastic
- ☐ Steel
- ☐ Nylon

8. How many control modes are integrated into the Pedro robot?

- ☐ 4
- ☐ 3
- ☐ 6
- ☐ None

9. What is the first step in 3D printing a part?

- ☐ Generate G-code
- ☐ Export to STL
- ☐ Design in CAD
- ☐ Print immediately

10. What makes Pedro unique in its construction?

- ☐ It is 100% 3D printed with no screws
- ☐ It uses wooden parts
- ☐ It is laser-cut
- ☐ It is glued together

11. What is an actuator?

- ☐ A device that converts energy into movement
- ☐ A type of sensor
- ☐ A computer screen
- ☐ A robot arm

12. What type of robot is Pedro?

- ☐ Industrial welding robot
- ☐ Educational robotic arm
- ☐ Medical robot
- ☐ Cleaning robot

13. Name one real-world use of robotic arms.

- ☐ Playing video games
- ☐ Assembling cars in factories
- ☐ Charging phones
- ☐ Watching movies

14. Why are robots useful?

- ☐ They can do dangerous or repetitive tasks
- ☐ They replace all humans
- ☐ They only entertain people
- ☐ They only work at night

15. What will you learn in the next lessons?

- ☐ How to program and control the robot
- ☐ How to build a car
- ☐ How to cook
- ☐ How to play music

16. Which communication method uses a cable?

- ☐ Bluetooth
- ☐ Radio
- ☐ USB Serial
- ☐ Wi-Fi

17. What happens in Record Mode?

- ☐ Pedro turns off
- ☐ Pedro stores the commands for later
- ☐ Pedro moves randomly
- ☐ Pedro charges its battery

18. Which modes allow Pedro to communicate remotely?

- ☐ Radio Mode
- ☐ Replay Mode
- ☐ Manual Mode
- ☐ Bluetooth Mode

Congratulations Mission Completed!

You have unlocked:  the Pedro Discover badge
Don't forget to stick your Pedro Badge into your Pedro Passport before starting the next lesson.

